# András Gémes

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## Summary

Security researcher with 8 years of experience. Hands-on experience in mobile application security, reverse engineering, malware analysis and vulnerability research. Certified in Sec+, CASP+/SecX, CEH and others. LLVM compiler engineering (code obfuscation) and embedded systems background. Open to roles in mobile security.

## Work experience

**Application Security Evaluator** @ *SGS Brightsight - Delft, Netherlands* **June 2026 – Present**

- Evaluating mobile payment applications against the PCI MPoC standard

**Security Researcher** @ *HighTec EDV-Systeme GmbH - Budapest, Hungary* **Feb 2023 – June 2026**

- Analyzed Windows, Linux and Android malware, writing detection rules and config extractors (Ghidra, Python)
- Implemented custom LLVM-based code obfuscation pass plugins based on malware research
- Evaluated the security impact of reported LLVM vulnerabilities as a member of the LLVM security group

**Application Security Engineer** @ *Knorr-Bremse - Budapest, Hungary* **May 2018 – Jan 2023**

- Implemented and evaluated static application security testing across embedded C codebases
- Resolved embedded systems vulnerabilities discovered through code review and AFL++ fuzzing
- Built iOS and Android apps for real-time vehicle data visualization

## Technical skills

**Reverse engineering (static):** DiE, Ghidra, IDA, ILSpy, dnSpy, Binwalk, Apktool, jadx, otool, ipsw
**Reverse engineering (dynamic)**: x64dbg, Sysinternals, LLDB, GDB, Frida, ADB, QEMU, Qiling, VirtualBox, strace
**Detection engineering:** YARA, Sigma, Suricata, capa
**Programming languages:** C, C++, Rust, Objective-C, Swift, Python 3, Java, Assembly (ARM64, x86-64), Bash
**Vulnerability research**: checksec, ROPgadget, AFL++
**Network and traffic analysis**: Wireshark, Burp Suite, mitmproxy, Zeek, FakeNet-NG, INetSim
**Platforms and DevOps tools:** Linux (Fedora, Ubuntu), macOS, Windows, Git, Docker, GitHub Actions, Jenkins

## Certifications

CompTIA Security+, CompTIA CASP+/SecurityX, EC-Council CEH, TCM Security PMAT, Invoke RE IMBT and others.

## Open source contributions

- ghidra: contributing bug reports and patches to Ghidra, focusing on the BSim, Debugger and FunctionID features
- threat-detection-rules: publishing YARA, Sigma and Suricata rules for analyzed malware families
- ghidra-scripts | reversing-scripts: developing custom scripts to support reverse engineering
- phantom-pass | o-mvll: implementing and improving LLVM-based code obfuscation pass plugins
- rust-arm64: writing a Rust book about analyzing Rust-to-ARM64 compilation

## Education

**MSc in Mechatronics Engineering** @ *Budapest University of Technology and Economics* **Feb 2016 – June 2018**

**BSc in Mechatronics Engineering** @ *University of Pannonia* **Sept 2012 – Jan 2016**

## Continuous education

Currently learning on Mobile Hacking Lab (Android and iOS application security courses).